CODE BUSTERS TEST

Clements Invitational

Name: _____

Team Name: _____

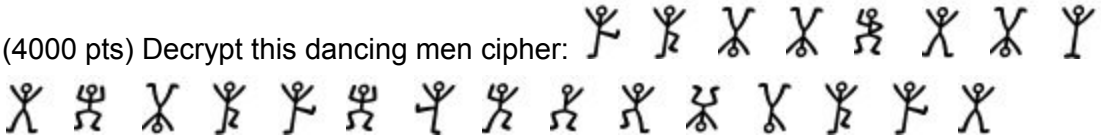
Team #: _____ Score: _____



"Do we really need to encrypt our data? Most of our communications are impossible to understand in the first place."

1. (2000 pts) Decrypt the following Vigenere cipher using the key ANTONY: I PHAR RO ONFL AARLOE, LOG MC CPAVLS UGM.

2. (5000 pts) Decrypt the following message using mono-alphabetic substitution: PUBOTGSSQFEUZEBDUFQ

3. (4000 pts) Decrypt this dancing men cipher: 

4. (4000 pts) Given the key of encryption is a= 5 and b= -2, decrypt this affine cipher: MK OQU FSYNO PQ IQGGMP PFSYKQL?

5. (2000 pts) Decrypt this monoalphabetic cipher: QHE CEG WP FWZHQWEM QHIG QHE PLUMX.

6. (6000 pts) Given that the following matrix was used as the encryption matrix, decrypt: NGCH EH XBOIKBVS YRZ INGRS.

$$\begin{pmatrix} H & O \\ W & L \end{pmatrix}$$

7. (4000 pts) Given the key: ETTUBRUTE decrypt this Vigenère Cipher: RHM NIRN B PSOXX DRYLEV EXMT, SOM XLTM C MFPXH VHFY NFLX.

8. (4000 pts) Using the following matrix for Hill encryption, encrypt: ANY FOOL CAN CRITICIZE, CONDEMN, COMPLAIN

$$\begin{pmatrix} B & U & T \\ W & H & Y \\ T & H & O \end{pmatrix}$$

9. (6000 pts) Decrypt the following message using mono-alphabetic substitution: TISM PO S YSL, PD IPO RIPXD UGGQ SLQ YSNZXM GD IPO MPYX WX WJM MG OAXXV SLQ DXXQ? S WXSOM, LG YGNX.

10. (3000 pts) Encode the following message using PINEAPPLES as the key for a Vigenère cipher: PIZZA IS A GREAT BLESSING

11. (6000 pts) Decrypt the following Affine cipher (hint a is a number divisible by 3 less than 20): KTEQ VTFEQ QSZ KRNB, PRLZ QSZ KRNB VTFEQ.

12. (5000 pts) Decrypt the following Affine cipher: TG JK TGC T SFC ZTY.

13. (3000 pts) Decrypt the following message using monoalphabetic substitution: EB NK BD IBE EB NK, EJSE HC EJK OQKCEHBI.

14. (6000 pts) Decrypt the following Affine cipher: IH’A E BYDDQ PYDDQ SBTIAHKEA!

Clements Invitational - Repaired practice edition

Division C | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

8. (2 points) Authored replacement for source item Question 8: What must be known before using a Baconian table?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

A. The most common English word

B. The paper color

C. The alphabet convention

D. The carrier sentence meaning only

Answer: _____